MIL-P-22968B(AS)
5 November 1984
SUPERSEDING
MIL-P-22968A(AS)
1 December 1966

MILITARY SPECIFICATION

PAPER, RECORDING, HEAT SENSITIVE

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification presents the requirements for one type of a heat-sensitive recording paper, in roll form, that furnishes permanent marking or tracing on one side.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-P-31	Paper; General Specifications and Methods of Testing
WW-T-700/2	Tube, Aluminum and Aluminum Alloy, Drawn, Seamless, General Specification for
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall
PPP-T-60	Tape, Pressure Sensitive, Packaging, Waterproof
PPP-T-76	Tape, Packaging, Paper (For Carton Sealing)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7530

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.

information to be analyzed.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Sleeves and Tubing-Interior Packaging
MIL-B-121	Barrier Material; Greaseproofed, Waterproofed, Flexible

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspections by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 985	45 Deg, 0-Deg Directional Reflectance For Blue Light (Brightness) of Paper
ASTM D 3951	Commercial Packaging

(Copies of ASTM standards or test methods may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of conflict between the text in this specification and references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. Materials shall consist of a suitably compounded, heat sensitive, coated paper containing no ground wood and free from any creases, splices or blemishes that will interfere with recording qualities. The coated paper shall be wound on an aluminum alloy, 3003 H-18, core. (See 4.4.1.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.

3.1.1 Rolls. A roll of paper shall be one continuous (no splices) strip 4.750 inches (maximum) in width and 115 +2 to -0 feet in length and wound on the core with the heat sensitive surface inside. Starting 20 feet ± 6 inches from the inside end of the roll, a black line 1/8 inch wide and 10 to 12 inches long shall be printed on one edge only of the heat sensitive surface. Starting 9 feet ± 6 inches from the inside end of the roll, a black line 1/8 inch wide and 10 to 12 inches long shall be printed on each edge of the heat sensitive surface. The national stock number shall be printed on the beginning edge of the outside of the roll. (See figure 1 for the markings on the roll.) (See 4.4.1.)

3.1.2 Core. The aluminum core shall be free from burrs, loose metal, sharp edges and shall conform to the dimensions shown in figure 2.

3.1.3 Coating. The coated side of the paper shall have a carbon black base coat evenly covered by a heat sensitive, waterproof, opaque-white coating with a whiteness equal to or greater than 68.4% reflectance. The coating may be of a cellulose nitrate derivative applied as a thin layer over the black substrate and so constituted that the heated stylus will produce a dark consistent line when the paper is used in Recorder RO-114/ASA-26 with the stylus in the center position of its travel. (See 4.6.1 and 6.1.)

3.1.4 Uniformity of spooling. The paper shall be wound in accordance with figure 1 at an even tension so as to prevent darkening of the paper due to excess tension and to prevent the aluminum core from being ejected due to insufficient tension under normal handling and running conditions. The paper shall be a continuous length free from splices, wrinkles, creases or weave which would interfere with its proper operation in the recorder. The paper shall not stick to the aluminum core so as to prevent the paper from unwinding freely. (See 4.4.1.)

3.2 Quality of recording. The quality of the recording shall be satisfactory when used in Recorder RO-114/ASA-26 or an acceptable substitute under normal operating conditions. The darkness (see 6.1) of the line shall be equal to 4.6% reflectance or darker at the center of the line. The width of the trace shall be 0.030 to 0.040 inch using a stylus width of 0.035 inch and stylus force on the paper of 8 grams. The stylus temperature shall be 300°C ± 15°C (572°F ± 59°F) and the paper speed shall be 0.6 inch per second. (See 4.6.1.)

3.3 Physical Characteristics.

3.3.1 Flammability. The coated paper shall not burn at a rate greater than 30 inches per minute when tested as indicated in 4.6.2.

3.3.2 Waterproofness. The coated side shall not turn darker than 59% reflectance when tested as indicated in 4.6.3.

3.3.3 Fumes resistance. The coated side shall not turn darker than 59% reflectance when tested as indicated in 4.6.4.

3.3.4 Heat resistance. The paper shall be capable of storage for 16 hours at 85°C (185°F) and 90% relative humidity and not turn darker than 59% reflectance and the coated side not become sticky. In addition the coated

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.

side shall not turn darker than 12% reflectance when exposed to an air temperature of 180°C (356°F). (See 4.6.5.)

3.3.5 Cold resistance. The coated side of the paper shall not delaminate or show crazing, cracking or flaking when exposed and tested as indicated in 4.6.6.

3.3.6 Finished paper.

3.3.6.1 Basis weight. The basis weight shall be 56g/m^2 ± 3g when weighed as indicated in 4.6.7.

3.3.6.2 Thickness. The thickness of the coated paper shall be .0020 to .0025 inch when measured as directed in 4.6.7.

3.3.6.3 Tensile strength. The tensile strength of the paper in the machine direction shall be 14 pounds minimum and in the cross machine direction shall be 7 pounds minimum when tested as indicated in 4.6.7.

3.3.6.4 Tear resistance. The tear resistance of the paper in the machine direction shall be 10 grams/cm minimum and in the cross machine direction 16 grams/cm minimum when tested as indicated in 4.6.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirments.

4.1.1 Inspection of materials and components. In accordance with 4.1, the contractor is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2 Classification of inspection. This specification provides for quality conformance inspection, which shall consist of the examinations and tests specified in 4.4 and 4.6.

4.3 Sampling for inspection. Sampling for inspection shall be performed in accordance with the provisions in MIL-STD-105. For purposes of sampling, an inspection lot shall consist of not more than 500 rolls of paper, made by the same process, from the same manufacturer and to be delivered at one time.

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:00Z
Check the source to verify that this is the currentversion before use.

4.4 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels and acceptable quality levels (AQL's) in 4.4.3. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of rolls for examination under 4.4.1 and in units of shipping containers for examination under 4.4.2.

4.4.1 Examination of the end item for defects in appearance, construction, identification and workmanship. The sample unit for this examination shall be one roll.

No.	Examine	Defects	Classification
101	Roll	Creases, splices or blemishes	Major
102	Coating	Not opaque-white	Major
103	Core	Burrs, loose metal, sharp edges, dented	Major
104	Tape	Missing	Major
105	Identification	Missing Not as specified	Major
106	End of roll marking	Missing Not as specified	Major
107	Spooling	Uneven tension, paper stuck to the core	Major

4.4.2 Examination of packaging. Examination shall be made to determine compliance with preservation, packing, and marking requirements of Section 5. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed.

No.	Examine	Defects	Classification
108	Material	Size of unit pack not as specified. Size of shipping container not as specified.	Major
109	Workmanship	Level of preservation and packing not as specified. Incomplete or insecure sealing of unit pack or shipping container flaps. Bulging or distortion of shipping container.	Major

Source: https://assist.dla.mil -- Downloaded: 202640-19T15:00Z
Check the source to verify that this is the current version before use.

109 (cont.)		Broken, bruised, crushed or otherwise damage shipping container. Marking of unit pack and shipping container not marked in accordance with MIL-STD-129.	Major
110	Weight	Gross weight in excess of weight limitations of container or unitized shipment of unit packs and intermediate packs incorrect.	Major

4.4.3 Inspection levels and acceptable quality levels for examination. The inspection levels for the purpose of determining the sample size and the acceptable quality levels (AQL's) shall be expressed in percent defective. The lot size shall not exceed 500 rolls.

Examination of paragraphs	Inspection level	AQL
4.4.1	II	1.5
4.4.2	S-4	6.5

4.5 Inspection level and acceptable quality level for tests. The inspection level for the purpose of determining the sample size and the acceptable quality level (AQL) shall be expressed in percent defective. The lot size shall not exceed 500 rolls.

Test paragraph	Inspections level	AQL
4.6	S-3	4.0

4.6 Tests. When specification requirements specify that test results are to be certain percentages of reflectance, ASTM D 985 is an acceptable reflectance test method to be used.

4.6.1 Quality of recording.

4.6.1 Procedure. A trace shall be produced on the recording paper using Recorder RO-114/ASA-26 (see 6.1) or an acceptable substitute under simulated normal operation. The width and reflectance of the trace and the speed of the paper shall be measured. The paper shall meet the requirements in 3.1.3 and 3.2.

4.6.2 Flammability.

4.6.2.1 Procedure. The test shall be conducted in a draft free atmosphere maintained at 50 ± 5% relative humidity and 23° ± 2°C (73° ± 3.5°F) temperature. A one inch strip of recording paper when held vertically and ignited so as to burn from the top downward shall not burn faster in this atmosphere than indicated in 3.3.1.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T15:00Z
Check the source to verify that this is the current version before use.

4.6.3 Waterproofness.

4.6.3.1 Procedure. Apply a drop of distilled water or a drop of sodium chloride solution to coated side of a two inch strip of recording paper and let stand 15 minutes at 15°C (59°F). The area under the drop shall not turn darker than that stated in 3.3.2.

4.6.4 Fumes resistance.

4.6.4.1 Procedure. In an ambient atmosphere of 50 ± 5% relative humidity and 23° ± 2°C (73° ± 35°F) temperature, place a strip of recording paper in the upper sections of a closed container that has liquid gasoline at the bottom. Expose the paper to the gasoline vapors for 15 minutes. The paper shall not turn darker than indicated in 3.3.3.

4.6.5 Heat resistance.

4.6.5.1 Procedure. After storage as stated in 3.3.4, the recording paper shall be tested as follows: (a) the reflectance of the paper determined for compliance with the value stated in 3.3.4 and (b) the smoothness of the coated side tested by unrolling the paper and allowing a two-foot length of paper to hang downward, and slowly rotating the core to determine the ease of movement. A specimen of test paper shall be exposed to an air temperature of 118°C (356°F), and the reflectance of the paper determined for compliance with the value stated in 3.3.4.

4.6.6 Cold resistance.

4.6.6.1 Procedure. The paper shall be tested as follows:

(a) Exposed for 16 hours at -62°C (-796°F) and examined for the defects stated in 3.3.5.
(b) Exposed for 8 hours at -30°C (-22°F). While at -30°C (22°F), pulled over a 90° steel corner having a radius of .0005 inches at a speed of 15mm per second, with a pull of one pound, the paper bent 90°. Examined for the defects stated in 3.3.5.

4.6.7 Finished paper. The finished paper shall be tested in accordance with the test methods in UU-P-31. (See 3.3.6 for requirements.)

Test	Method
Basis	110
Thickness	173
Tensile Strength	171
Tear resistance	170

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Level A preservation shall be Method 1C-1 in accordance with MIL-P-116.

5.1.1.1 Unit packing. Each roll shall have the outer flap of paper secured with a one inch strip of pressure sensitive tape, conforming to PPP-T-76, to prevent unwinding. Each roll shall be wrapped in barrier material conforming to type I, grade A of MIL-B-121 or sealed in a barrier bag conforming to type I, class E of MIL-B-117. The wrapping procedure, where used, shall completely cover the roll and have all loose edges secured with pressure sensitive tape, conforming to PPP-T-76. (See 4.4.2.)

5.1.1.2 Intermediate packing. The specified number of unit packs, prepared as specified in 5.1.1.1, shall be placed in intermediate containers conforming to type CF or SF, class domestic of PPP-B-636 and closed in accordance with the container specification (see 6.2). The gross weight of the intermediate container shall not exceed 20 pounds. (See 4.4.2.)

5.1.2 Commercial. Rolls shall be preserved in accordance with ASTM D 3951.

5.1.2.1 Intermediate packing. The specified number of unit packs prepared as specified in 5.1.2 shall be placed in intermediate containers conforming to ASTM D 3951 (see 6.2).

5.2 Packing. Shipping containers shall be of minimum cube consistent with the level of protection specified. All shipping containers will contain the same number of intermediate containers snugly packed. The levels of packing shall be A, B or commercial.

5.2.1 Level A. The specified number of intermediate packs shall be packed in overseas containers conforming to PPP-B-621 or PPP-B-601 (see 6.2). Strapping and closures shall be in accordance with the appendix of the applicable shipping container. The gross weight of the shipping container shall not exceed 70 pounds. (see 4.4.2 and 6.2)

5.2.2 Level B. The specified number of intermediate packs shall be packed in weather resistant containers conforming to PPP-B-636 or PPP-B-640 (see 6.2). Closures shall be in accordance with the appendix of the applicable container specification. The gross weight of the shipping container shall not exceed 70 pounds. (See 4.4.2 and 6.2.)

5.2.3 Commercial. The specified number of intermediate packs shall be packed in shipping containers in accordance with ASTM D 3951. (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Military. Each unit pack, intermediate pack and shipping container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Each unit pack, intermediate pack and shipping container shall be marked in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The heat sensitive recording paper in this specification is intended for use in RO-114B/ASA-26, Recorder, Signal Data part of Recorder Group AN/ASA-26A.

6.2 Ordering data. Acquisition documents shall specify the following:

(a) Title, number and date of this specification.

(b) Level of preservation and packing required.

(c) Quantity of unit packs per intermediate container. (See 5.2.)

(d) Quantity of intermediate containers per shipping container. (See 5.3.)

(e) Special markings, if required.

Preparing Activity:

Navy-AS

Project No. 7530-N196

Source: https://assist.dla.mil -- Downloaded: 202640-19T15:00Z
Check the source to verify that this is the currentversion before use.

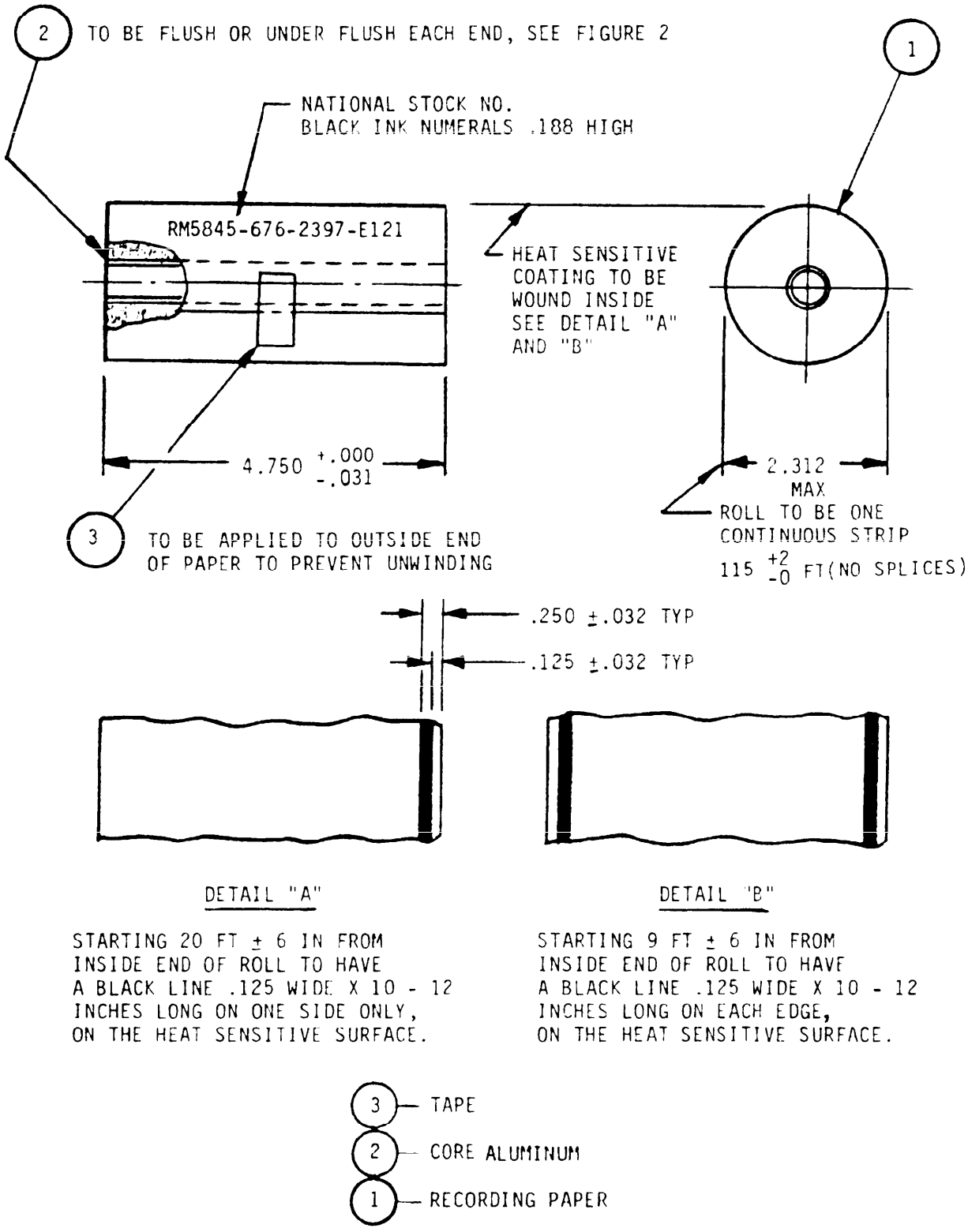
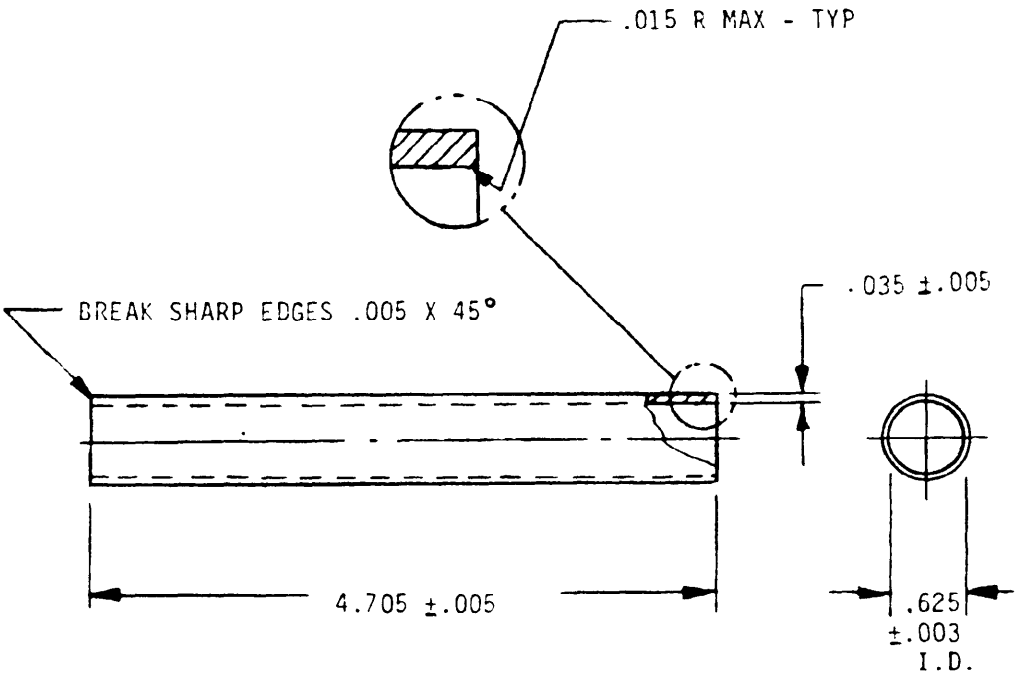


Figure 1. Dimensions and markings of recording paper for use in Recorder RO-114A/ASA-26.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.



MATERIAL:

ALUMINIUM ALLOY 3003, H-18, PER WW-T-700/2

Figure 2. Dimensions of aluminum core

Source: https://assist.dla.mil -- Downloaded: 202610-19T15:00Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(FSSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON, D.C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(FSSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-22968B(AS)	PAPER, RECORDING, HEAT SENSITIVE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:00Z
Check the source to verify that this is the current version before use.